

Phases 3, 4 and 5: what was built

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All three phases are on the server. This is what each one does, what changed after your testing, and what the game itself will not allow.

Your build plan was also read back line by line against what was running. Five things it listed for phases 1 to 5 had never been written: Team, experience and levels, kick and ban and granting XP, records that split by class, and a qualifying session that sets the grid. They are written now and they have a section of their own near the end. One of them, Team, I had wrongly told you was covered by the race lobby. It was not, and it is built properly now.

1 Phase 3: the bottom bar

The five buttons work. The server decides whether you may press one, holds the car still for as long as the job takes, and adds the penalty. The game does the work.

Button	Hold	Penalty	What it does
Reposition	5s	60s	Back onto the course, facing the way it runs. Section 2.
Spare tire	30s	30s	A tire off the rack. Section 3.
Repair	60s	30s	The whole car rebuilt, in place.
Fuel +25%	20s	none	A quarter of a tank, or a full one in the pit.
Re-rack	30s	none	Pit only. Fills the spare rack again.
Lights	none	none	Headlights, fog, hazards, lightbar, siren, horn, flash.

The penalty lands when the hold starts, not when it ends, so quitting half way through buys nothing. A job the game cannot do gives the seconds back. Outside a run everything is instant and free. Every number lives on the server and can change without anyone reinstalling.

Each job makes a noise while the car is held, from the files you sent, and only while a race is on. Spare tire is the impact driver six times, one per nut. Fuel is the bottle at four fifths speed. Repair drills once and then hammers and ratchets until the hold runs out, so it follows the hold rather than a fixed count. Reposition and Re-rack are quiet, as you asked.

2 Reposition

Standing the car up where it rolled leaves it nose into whatever stopped it, which is not what anyone means by the word. It goes back onto the leg you are driving instead, the line from the gate you last passed to the one you are going for, at the nearest point on that leg and pointing the way the course runs.

Nearest is the fair one. No ground is given and none is taken, so it cannot be used to skip a corner. The damage still comes with you. Outside a race it stands the car up where it is, as before.

3 The spare tire

Your idea was the right one, and better than what was there before. Spare tires are real parts bolted to the car, so the rack is read off the vehicle rather than guessed at. A race truck carries

two, a UTV one, plenty of things none. That is the limit, not a number in a settings file, and a car with no rack has no spare button at all.

Which road the button takes depends on how bad the tire is.

- Flat with the wheel still under it: the air goes back in where you stopped and nothing else on the car is touched. This is most racing flats.
- Shredded, or the wheel is broken: the tires are replaced out of the parts list, the same job the vehicle selector does. It is the only thing the game can do for a tire that far gone.

Either way one comes off the rack. The button shows how many you have left and greys out at zero. Re-rack is a pit job: it fills the rack again for the wait and nothing on the clock.

One thing to know, because it decided the design. Replacing a part rebuilds the whole car, not just the tires. So that road is only taken when the tire is genuinely destroyed. Run it on every flat and each puncture hands out a free repair, and nobody presses Repair again, since Repair takes twice as long for the same penalty. The cost of doing it this way is that on an ordinary flat the spares on the rack still look full on the model even though your count went down. The count is what the rules run on.

4 Phase 4: races you make and join

The touchy motion start is gone. A race is a thing you make, and others join.

1. Pick a course, laps and mode, and make the race **public** or **invite only**, choosing people from the player list.
2. Everyone who joins is placed on the grid. Nothing is armed. The start line does nothing, however many times it is crossed.
3. The host presses **Start Race**. Only then does the line go live.
4. Each driver's clock starts when they cross it, so the timing stays exact for everyone even if they roll off a second apart.

Only the host can start it. A race that has started refuses late joiners. If the host leaves before starting, the race passes to whoever joined first rather than dying. One race per course, different courses at the same time are fine, and solo practice works exactly as before.

5 Phase 4: the pit and the speed zones

A checkpoint can be marked as a **pit** when a course is built. It is not part of the lap, it is drawn in its own colour, and while you are inside it:

- Fuel fills to 100% instead of a quarter.
- Repair and Spare tire add nothing to your race time.
- The holds still run in full, and the clock is still going. A stop costs the time it takes and nothing on top.

Speed zones are the 37 mph idea, between any two checkpoints you pick. A zone is a start gate, an end gate and a limit, and a course can have as many as needed. The server reads the speeds it already collects, so nothing is trusted to a driver's own game. Going over draws a warning first; the 30 second penalty lands only after three seconds over, so one bump over a crest is free, and it will not charge twice until you have dropped back under.

6 Phase 5: the record books

Every course keeps a book per mode. The Records button shows them.

- Each driver's best corrected run, fastest first, with when it was set.
- The fastest clean lap anyone has driven, held separately.
- Your own position when you are outside the top ten.
- Controller and wheel are separate books, as agreed in phase 2.

The moment a race finishes, the run goes into the books before the results go out, so the results screen hangs the badge on the row it belongs to: **course record** or **personal best**. The books live on disk and survive restarts.

What never makes the books: a run the server marked as suspect, because its clock could not be vouched for, and any lap with a cut checkpoint in it, which was already refused a best lap by the results.

7 Fixed along the way

The recordings you sent found real bugs, and each one led somewhere useful.

What you saw	What it was
Checkpoints only counted near the middle of the road, then later did not span it at all.	A gate is centred on wherever the capture car drove. Rays fired sideways to find the edge touch nothing on an open dirt track, so a gate now takes its width and its middle from the road the game's own driving ai uses. That is measured rather than guessed. A road narrower than the width you set leaves your width alone.
Reposition put the car back where it had rolled to.	It goes back onto the leg being driven now, at the nearest point on it. See section 2.
The race picker opened with its heading cut off and a bar down the side.	A window could be dragged shorter than what was in it, and the size was remembered. It cannot be now, and a size already saved heals itself the next time the window opens.
A window opened when a race started, and had to be closed to drive.	Nothing is drawn over the road before the clock exists. A message says the race is on and counts itself out.
Some checkpoints floated and cars passed under them.	The volume was centred on its base. It now stands from just below the ground to its full height.
Missing one checkpoint froze the ones after it.	Several causes, all gone: the volume was a slot along the road instead of a gate across it, and a run that skipped too many gates was refused forever.
The clock read 4:60.0.	The minutes were split off before the rounding.

What you saw	What it was
Repair teleported the car across the map.	A bare engine reset means "back to spawn". Both movers now place the car exactly where it stands.
The clock jumped backwards thirty seconds.	A checkpoint charged as cut had been reached after all, and the penalty came back off. Correct, but silent; it now says so on screen.
The light menu would not close.	It closes on the light you picked, on reaching past it for another button, and on driving off, which is the one a click never covered. Messages run a ten second line and go.
The spare tire button killed the interface.	New code was written above the part that calls it, which lua does not allow. It parses and then dies on the first press. A build check now reads every client file for that mistake, and another follows every button through to the code behind it, which caught a second button that did nothing at all.

8 The numbers

Suite	Tests
Gate geometry and putting back	30
Course capture and shape	31
Phase 1, identity	91
Phase 2, racing	131
Phase 3, the bottom bar and the rack	68
Phase 4, lobby, pit, zones	63
Phase 5, the books	31
Teams, xp, bans, classes, qualifying	84
Results	120
The clock and lap bar	26
Total	675

Plus 190 interface checks, 10 key bindings, and a reader over every client file that fails the build when a name is used above where it is written. All of those fail the build when the screen and the code stop agreeing.

The books, the bar, the rack and the gate sums were also tested by breaking them on purpose: charging at the wrong end, letting marked runs into the books, never taking a spare off the rack, re-racking out on the course, ignoring the road width, putting a car back where it already was, ignoring the class a driver entered, letting a made up class through, keeping one best per course instead of one per class, dropping the class off the line your bot reads, and starting a grid without the class its race was made in. Every sabotage was caught, which is what says the numbers above mean something.

9 What the game will not allow

Written down so nobody goes looking for a setting that does not exist.

- One tire cannot be mended on its own. Air can go back in, which covers punctures. A destroyed one can only come back by rebuilding the car from its parts, which is why the spare works the way section 3 describes.
- A car cannot be frozen for one player only in a way other players do not see. The hold freezes the car for everyone, which is correct for a race anyway.
- Gate widths come from the road the driving ai uses. Your dirt oval has one. A course built somewhere with no roads defined falls back to the width you set in the capture tool.

10 The rest of what the plan promised

Your build plan was read back line by line against what was on the server. Five things it listed for phases 1 to 5 had not been written. They are written now, and this is what each one does.

Team

Invite somebody into your car, or ask to get into theirs. You both have to be in the same vehicle model: it is checked when the offer goes out and again when it is taken up, because cars change while an offer sits waiting. Swap to another car afterwards and the team ends on its own and says why. So does one of you leaving the server.

The results screen keeps both of your rows, because that is where the splits and the penalties live, and puts a team standing beside them. Both of you have to finish, and the team's time is your two corrected times added together, so a team is only as quick as the pair of you.

Experience and levels

Finishing pays by where you came: two hundred for the win, two less for each place after it, never below twenty. That is your ladder, and the numbers are in the settings file rather than the code, so they move without anyone reinstalling. A level is a thousand points.

A run the clock could not vouch for pays nothing, the same rule the record books use. Guests earn nothing either, having no record on disk to put it in. XP sits beside the name, so it survives a restart for the same reason the name does.

Kick, ban and granting XP

Four commands in the console: `rm kick`, `rm ban`, `rm unban` and `rm xp`. There is a `rm bans` to see who is on the list and why.

A ban follows the account rather than a name, which can be changed, or a session, which is handed out fresh every join. It is on disk, so a restart does not quietly let everybody back in. Somebody banned is turned away as they knock, before anything is written down for them. In game nobody can act on somebody who ranks as high as they do, or on themselves.

Records by class

Your seventeen classes are in, as you wrote them, in your order, under the two divisions you grouped them under. They are on the race screen as a row of buttons under Limited and Unlimited.

A class is entered rather than worked out from the car. You allow every vehicle, and two drivers in the same truck can be running different classes, so the car cannot tell the server which book a run belongs in. The driver picks one before they set off, or whoever makes the race picks one for the whole grid, which is the way your bot announces them. Picking nothing is allowed: the run still counts, it just sits outside the class books.

Your best is kept once per class, not once per course. A slower run in a second class is that class's first entry rather than a run that failed to beat you. Top of your own class earns a record of its own on the results screen, beside the outright course record.

The record screen has a filter across the top: every class, or one of them. It only offers the classes that have actually been raced on that course, so it stays short.

One thing to check. Your list had a bullet reading Class 5/1600, 1600 and Class 9. I put that in as one class exactly as you wrote it. If those are meant to be three separate books, say so and I will split them.

A class can also be pinned to particular vehicles later, which would refuse the wrong car at the line. Nothing is pinned today, because you said every vehicle is allowed for now.

Qualifying

A course marked as qualifying now sets the grid. Whoever was quickest in the qualifying session starts at the front of the next race on that course, and anyone who was not in it lines up behind those who were, in the order they joined, so a late arrival starts at the back rather than being turned away. The order lasts the session: a grid earned this evening should not still be deciding places next week.

One difference from the plan, named rather than hidden

Speed zones were written as holding the car to the limit. What is built reads your speed on the server, warns you, and charges thirty seconds if you stay over for three. It does not physically slow the car. Enforcing by penalty cannot be cheated by a driver's own game, which is why it went that way, but it is not the same thing as a cap and you should know that.

11 Still open

- The look of the interface, which you asked about. It is a separate job and can start whenever you like.
- CoPilot and Challenges are phase 6.

Everything described here is on the server, sounds and all. The build is **0.5.6-classes**, and the console says so on every restart, so you can always tell which one you are running.

The record books start empty, so the first finished race on each course sets its first record. Experience starts at nothing too: levels only move from here on, because there are no old results to pay out for.